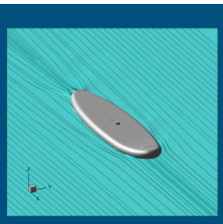
For our Fluid Mechanical Dissection, our group analyzed a manual air pump for an inflatable paddle board, a two-cylinder design with a switch that lets you trade off high volume/low pressure vs high pressure near the end, along with a pressure gauge in the handle.

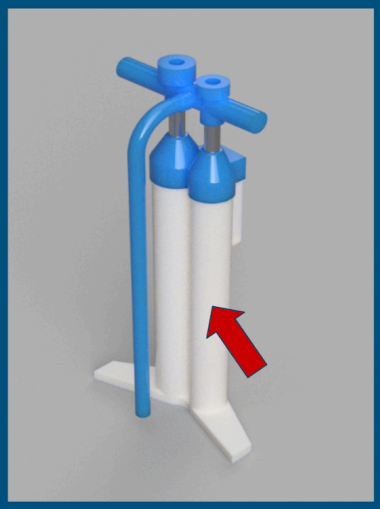
My contribution focused on the parametrization and assumptions behind our analysis. I identified the key non-dimensional parameters (Reynolds number and Mach number) for the air flow in the 25 mm diameter, 1.5 m hose, set the scaling (using hose diameter for length and average hose speed (U = Q/A) for velocity), and used those to justify treating the hose flow as turbulent and the air as effectively incompressible for our calculations. I also documented the assumptions that we ran calculations in Mode 3 (small tube open/big tube closed) and modeled the tubes as perfect cylinders.

Please check out my group's [Fluid Mechanical Dissection Presentation]!

Parametrization



- Governing parameters: Reynolds number (Re) and Mach number (Ma)
- Flow: air in the 25 mm diameter, 1.5 m flexible hose from the barrel
- **Scale: length = hose inner diameter, velocity = average hose speed U = Q / A**
- Re using (ρ, U, D, μ) > 4000 → turbulent hose flow
- Ma = U / speed of sound ≈ 0.02 → air is effectively incompressible
- Same Re and Ma definitions for Modes 1-3 → changing mode only changes U and Re as well

Assumptions

- We are doing calculations assuming we are using the mode 3, when the small tubes is open and big tube is closed to pump the paddle board
- We are also assuming the tubes as perfectly cylindrical tubes for the ease of calculation and analysis